Guided Inquiry: Exploring Cubic Equations of State

Thermodynamics Lab

May 31, 2026

Objective

To understand the strengths, limitations, and physical reasoning behind the van der Waals (vdW), Soave-Redlich-Kwong (SRK), and Peng-Robinson (PR) cubic equations of state by comparing their calculations for substances with diverse chemical attributes.

Background

Cubic equations of state are standard models in chemical engineering for predicting real-fluid behaviors. While the Ideal Gas Law ($PV_m = RT$) is simple, it ignores intermolecular attraction and molecular volume. Cubic EOS models introduce corrections for these non-idealities:

- **van der Waals (vdW):** The simplest modification, utilizing parameters a (molecular attraction) and b (molecular excluded volume).
- **Soave-Redlich-Kwong (SRK):** Introduces a temperature-dependent attraction parameter $\alpha(T, \omega)$ making use of the *acentric factor* (ω) to capture molecular non-sphericity.
- **Peng-Robinson (PR):** Modifies the attractive pressure term to yield substantially improved liquid density predictions, particularly for hydrocarbons.

Chemical Species Studied in This Activity

Methane (CH_4)	**Carbon Dioxide (CO_2)**	**Water (H_2O)**	**Pentane (C_5H_{12})**
Small, Non-polar	Non-polar, Quadrupolar	Polar, Hydrogen-Bonding	Large Hydrocarbon Chain
$\omega = 0.011$	$\omega = 0.224$	$\omega = 0.344$	$\omega = 0.251$

Figure 1: Overview of the molecular characteristics and acentric factors (ω) of the selected species.

Guided Inquiry Exercises

Part 1: Baseline Behavior — Methane (CH_4)

Methane is a small, symmetric, non-polar molecule. It exhibits weak dispersion forces, serving as a clean baseline for evaluating real gas behaviors.

1. **Ideal Conditions:** Set the species to **Methane**. Adjust the sliders to a temperature far above its critical temperature ($T > 2T_c$) and a low pressure ($P < 0.1P_c$).
   (a) Record the molar volumes computed by all three models. Do they agree?
   (b) Explain why all three models merge toward a single value under these conditions using molecular-level theory.

2. **Near-Critical Behavior:** Move both temperature and pressure sliders to near Methane's critical properties ($T_r \approx 1.0$, $P_r \approx 1.0$).
   (a) What happens to the distinct difference between the computed liquid and vapor volumes as you approach the critical point?
   (b) Do you observe significant divergence between the three equations of state here?

3. **Two-Phase Coexistence:** Keep pressure moderate ($P_r \approx 0.5$) but slide the temperature down to $T_r \approx 0.85$. The system should report a "Two Phase (L-V)" status.
   (a) Record the liquid molar volume (V_l) and vapor molar volume (V_v) predicted by the three models.
   (b) Which equation predicts the smallest liquid molar volume? Why does this matter when designing liquid storage units?

Part 2: Molecular Size and Chain-like Complexity — Pentane (C_5H_{12})

Pentane is an unbranched hydrocarbon. Its chain-like structure means it is not spherical, resulting in a significantly larger acentric factor.

4. **The Impact of Shape (ω):** Locate Pentane's acentric factor (ω) on the interface.
   (a) How does its acentric factor compare to Methane's?
   (b) Given the formulas for SRK and PR displayed on the screen, how does a non-zero ω directly affect the calculations of the attractive parameter $a(T, \omega)$?

5. **Corresponding States Comparison:**
   (a) Set Methane's sliders to $T_r = 0.90$ and $P_r = 0.5$. Record the liquid molar volume from the Peng-Robinson (PR) model.
   (b) Switch to Pentane and adjust its sliders to the exact same reduced states ($T_r = 0.90, P_r = 0.5$). Record its PR liquid molar volume.
   (c) Are the absolute molar volumes of Methane and Pentane similar at identical reduced states? What physical properties explain this difference?

Part 3: Highly Polar Systems — Water (H_2O)

Water molecules are highly polar and form strong, directional hydrogen bonds.

6. **Hydrogen Bonding Limits:** Select **Water** and move the sliders into the two-phase region (e.g., $T_r = 0.95, P_r = 0.8$).
   (a) Compare the liquid volumes computed by all three equations. Do they deviate more or less from each other than they did with Methane?
   (b) Classic cubic equations of state do not explicitly contain a term for directional hydrogen bonding. Knowing this, how reliable are these models for highly polar liquids?

Part 4: Synthesis & Conceptual Check

7. **Structural Complexity:** Rank the three equations of state in order of increasing complexity. Identify the specific terms or variables present in the advanced equations that are absent in the van der Waals model.

8. **Application Decision Matrix:** Fill out the following table indicating which EOS you would choose for each engineering scenario, citing a physical or mathematical reason:

Engineering Scenario	Preferred EOS	Thermodynamic Justification
Sizing an industrial separator for a mixture of $C_2 - C_5$ hydrocarbons.		
Quick, back-of-the-envelope estimation for a simple, non-polar gas far from its critical point.		
Saturated water-vapor phase equilibrium calculations for a steam power cycle.		